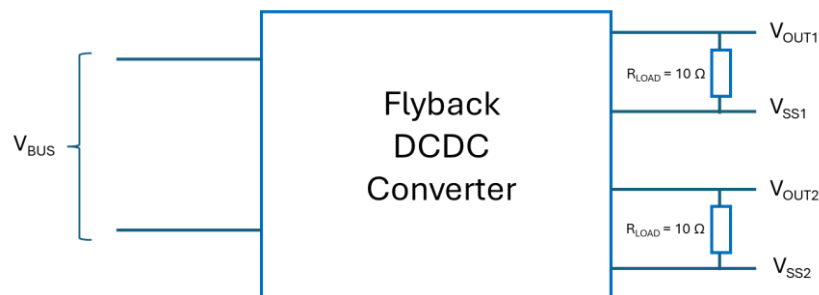


- The writing quality of your assignment report will affect your grade. If you used any references, you must include them at the end of your report. Moreover, your assignment should have a cover page.

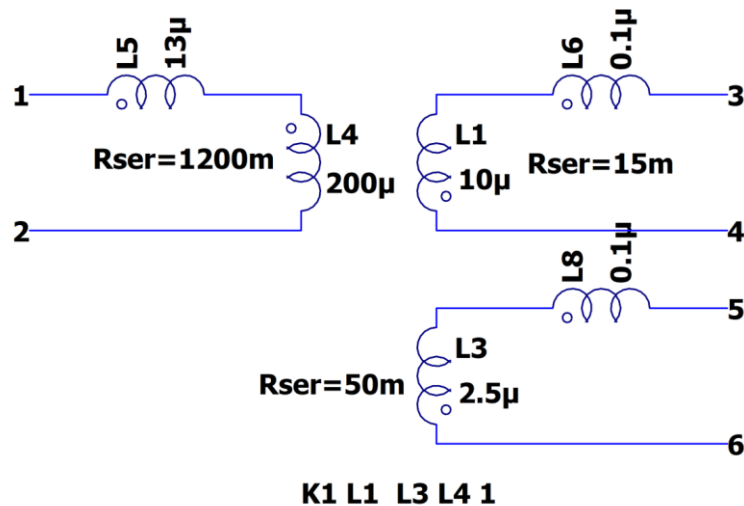
Question [100 pt]

Design a Flyback DCDC Converter which an input voltage V_{BUS} to a lower voltage V_{OUT1} and V_{OUT2} . The Flyback DCDC shall fulfill the following specifications:

- V_{BUS} is between 300 V and 500 V [10 pt].
- $V_{OUT1} = 15$ V and $V_{OUT2} = -7$ V and a resistor of $10\ \Omega$ is connected to both outputs [30 pt].
- Maximum output voltage ripple = ± 200 mV [20 pt].
- Minimum Efficiency = 75% [30 pt].
- The Flyback DCDC converter shall have an isolated feedback path to regulate the output voltages [10 pt].



The transformer can be designed as follows:



L5, L6 and L8 are the stray inductances of the transformer, therefore you should also design a snubber circuit to suppress voltage peaks at the drain of your transistor (please investigate the lecture slides for the Flyback DCDC converter topology, stray inductances and snubber circuit).

These voltage peaks shall not exceed the maximum rated drain-source voltage of your MOSFET!

You must make a simple thermal analysis (by simulations and/or calculations) of the components. The maximum rated junction temperature of the components shall not be exceeded. The simulated junction temperature of the MOSFET, snubber circuit components and the diodes shall be included in the report.

The submission shall include the LTSpice simulation file including all the additional models. Your simulation report shall include all the simulation results that prove that your Flyback DCDC converter fulfills the given specifications (input voltage range, voltage ripple etc.).

Hint: You can use the DCDC controller LT1243 from Analog Devices (can be found directly in LTSpice library) and the zip file *simulation_package.zip*, which includes a simple transformer and a transistor model.

Important Notes:

- 1) You must submit a single report in pdf format with a proper cover page. The report should include your methodology, circuit, simulation results and the interpretation of your simulation results.
- 2) In addition to the report, you must also submit the LTspice schematic file (.asc) of your Flyback DCDC Converter **with all the models** that you have used in your design!
- 3) You are expected to complete all work individually. **This is not a group project.** Any form of copying or academic dishonesty will result in grade reduction.
- 4) **If you have questions about the project, they will only be addressed during lecture hours. You are expected to complete this project using the fundamental concepts covered in lectures. In addition, you are expected to conduct your own literature research to support your approach and design choices.**